By: Ana Júlia B. Bezerra

Least Square Method

The least squares method is a statistical technique used to determine the best-fitting line or curve to a set of data points by minimizing the sum of the squares of the vertical deviations (residuals) from each data point to the line or curve. For a polynomial of degree 1, this method finds the best-fitting linear equation of the form $y = ax + b$.

Steps to solve a problem given the data points:

1. Set up the normal equations:
   - The linear equation is $y = ax + b$.
   - To find the coefficients a and b that minimize the sum of the squared residuals is necessary to use the following normal equations:

$$a \cdot S_x + b \cdot n = S_y$$

$$a \cdot S_{xx} + b \cdot S_x = S_{xy}$$

2. Calculate the sums needed:

$$S_x = \sum x_i, \quad S_y = \sum y_i, \quad S_{xx} = \sum x_i^2, \quad S_{xy} = \sum x_i y_i$$

   Where:

   S_x and S_y are the sums of the x-values and y-values, respectively.

   S_{xx} is the sum of the squares of the x-values.

   S_{xy} is the sum of the product of the x and y values.

3. Insert the values and solve for a and b.

Problem Consider the table:

x	2	2.5	4
$f(x)$	0.5	0.4	0.25

Using the least squares method, find the polynomial of degree 1.

1. Given the data points:

$$(x_1, y_1) = (2, 0.5)$$
$$(x_2, y_2) = (2.5, 0.4)$$
$$(x_3, y_3) = (4, 0.25)$$

2. Calculate the required sums:

$$S_x = 2 + 2.5 + 4 = 8.5$$

$$S_y = 0.5 + 0.4 + 0.25 = 1.15$$
$$S_{xx} = 2^2 + 2.5^2 + 4^2 = 4 + 6.25 + 16 = 26.25$$
$$S_{xy} = (2 \cdot 0.5) + (2.5 \cdot 0.4) + (4 \cdot 0.25) = 1 + 1 + 1 = 3$$

3. Set up the normal equations:

$$n = 3 \quad \text{(number of data points)}$$

$$a \cdot S_x + b \cdot n = S_y$$
$$a \cdot S_{xx} + b \cdot S_x = S_{xy}$$

Substitute the values:

$$8.5a + 3b = 1.15 \quad (1)$$
$$26.25a + 8.5b = 3 \quad (2)$$

4. Solve the system of equations:

From (1):

$$b = \frac{1.15 - 8.5a}{3}$$

Substitute b in (2):

$$26.25a + 8.5\left(\frac{1.15 - 8.5a}{3}\right) = 3$$

$$26.25a + \frac{8.5 \cdot 1.15 - 8.5 \cdot 8.5a}{3} = 3$$

$$26.25a + \frac{9.775 - 72.25a}{3} = 3$$

$$26.25a + 3.2583 - 24.0833a = 3$$

$$2.1667a + 3.2583 = 3$$

$$2.1667a = -0.2583$$

$$a \approx -0.1192$$

Substitute a back into the equation for b:

$$b = \frac{1.15 - 8.5(-0.1192)}{3}$$

$$b = \frac{1.15 + 1.0132}{3}$$

$$b \approx 0.7211$$

The polynomial of degree 1 $P_1(x)$ that best fits the given data using the least squares method is:

$$y = ax + b$$

$$P_1(x) = -0.1192x + 0.7211$$